

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level						
Sem:		I				
Paper Category:		DSC3-1 (Major) (Group - I)				
Paper Name:		Digital Electronics and Microprocessors			Paper Code : G06-0103	
Credit:		02		Theory:		2 Hrs./Week
Marks:	UA:	30	CA:	20	Total:	50

Course Objective:

1. To learn Boolean algebra and logic gates
2. To study digital logic families and their important features
3. To develop a designing and analyzing attitude about sequential circuits
4. To develop a designing and analyzing attitude about combinational circuits
5. To learn 8085 Microprocessor Architecture and Assembly language Programming.

Course Outcomes:

Upon successful completion of this course, students will be able to

1. Design and construct logic as well as arithmetical circuits
2. Calculate various important parameters of Digital logic families
3. Design & analyze combinational logic circuits
4. Design & analyze sequential logic circuits
5. To Execute 8085 Microprocessor Assembly language programming.

Unit-I:Logic Gates, Combinational and Sequential circuits:	No. of Lectures:15	Weightage: 12-18 Marks
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Logic Gates and Combinational Logic circuits: Introduction to logic gates, OR, AND, NOT, NAND, NOR, XOR, XNOR, Pin function of IC 7432, 7408, 7404, 7400, 7402, 7486 Applications of Logic Gates.

Combinational circuit: Introduction to the combinational circuit, Half adder, full adder, Half subtractor, Multiplexer (4:1), Demultiplexer (1:4), Encoder (4:2), Decoder (2:4), Applications of Combinational Logic Circuits.

Sequential circuits: Concept of sequential circuits, Flip-flops: RS, Clocked RS, JK, Master Slave JK, D Flip-flop, Pin configuration of IC-7474, Counter-synchronous, asynchronous (3-bit up counter and Down Counter), modulus of Counter (Mod 2, Mod 5, Mod 10), Pin configuration of IC 7490, Shift register (SISO, SIPO, PIPO, PISO) Pin Configuration of IC 7495), Applications of Sequential Logic Circuits.

Unit-II: Introduction to Microprocessor:	No. of Lectures:15	Weightage: 12-18 Marks
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Fundamentals of 8085 Microprocessor: Introduction to microprocessor, Basic system bus architecture, Concept of T state Machine cycle, Instruction cycle, pin function of 8085 microprocessor, internal architecture of 8085 microprocessor.

Instruction set of 8085: instructions Format, Classification of instruction set, Addressing modes, Assembly language programming of 8085(addition, subtraction, division, multiplication) Intel 8085 microprocessor features

Reference Books:

1. Morris Mano Computer System Architecture (3rd Edition) PHP
2. Digital principle & applications – Malvino Leech.3
3. Digital principle – Floyed.
4. Digital electronics – C. F. Strangio
5. Microprocessor Architecture programming and Application,-Ramesh Gaonkar